

THE CHOW RANK OF THE THREE-BY-THREE PERMANENT

ABSTRACT. We prove over every field of characteristic zero that

$$\text{ChowRank}(\text{perm}_3) = 4.$$

The lower bound follows from the kernel of the catalecticant–Koszul flattening, and the upper bound is Glynn’s four-term decomposition. The argument is self-contained, and an independent exact reconstruction program is included with the paper.

1. THE THREE-BY-THREE PERMANENT

1.1. **Definitions and statement.** Let

$$V = \text{span}_{\mathbb{C}}\{x_{ij} : 0 \leq i, j \leq 2\}, \quad \dim V = 9,$$

and define

$$\text{perm}_3 = \sum_{\sigma \in S_3} x_{0,\sigma(0)} x_{1,\sigma(1)} x_{2,\sigma(2)} \in \text{Sym}^3 V.$$

Definition 1.1. The Chow rank of a cubic form $f \in \text{Sym}^3 V$ is the least integer r for which

$$f = \sum_{s=1}^r \ell_{s1} \ell_{s2} \ell_{s3}, \quad \ell_{st} \in V.$$

Theorem 1.2.

$$\boxed{\text{ChowRank}(\text{perm}_3) = 4}.$$

1.2. **The catalecticant and Koszul flattening.** For a cubic form $f \in \text{Sym}^3 V$, define

$$C_{1,2}(f) : V^* \longrightarrow \text{Sym}^2 V, \quad D \longmapsto D(f),$$

and write $E_f := \text{im } C_{1,2}(f)$. Define

$$\delta : \text{Sym}^2 V \otimes V \longrightarrow V \otimes \bigwedge^2 V$$

by

$$\delta((uv) \otimes w) = v \otimes (u \wedge w) + u \otimes (v \wedge w), \tag{1}$$

extended linearly. The associated Koszul flattening is

$$K(f) := \delta \circ (C_{1,2}(f) \otimes \text{id}_V),$$

so

$$\text{im } K(f) = \delta(E_f \otimes V). \tag{2}$$

1.2.1. *The Koszul kernel and first prolongation.* For $E \subseteq \text{Sym}^2 V$, define

$$E^{(1)} := \left\{ g \in \text{Sym}^3 V : \frac{\partial g}{\partial x_{ij}} \in E \text{ for every } i, j \right\}.$$

Lemma 1.3. *There is a natural isomorphism*

$$\ker(\delta|_{E \otimes V}) \simeq E^{(1)}.$$

Consequently,

$$\text{rank } \delta(E \otimes V) = 9 \dim E - \dim E^{(1)}.$$

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Proof. Define

$$\iota : \text{Sym}^3 V \longrightarrow \text{Sym}^2 V \otimes V, \quad \iota(g) = \frac{1}{3} \sum_{i,j} \frac{\partial g}{\partial x_{ij}} \otimes x_{ij}.$$

Euler's identity shows that tensor multiplication sends $\iota(g)$ to g , so ι is injective. Applying (1) to a cubic monomial shows that the wedge terms cancel in pairs; hence $\delta \circ \iota = 0$.

Conversely, write the coefficients of $z \in \text{Sym}^2 V \otimes V$ as z_{abc} . Symmetry in the first two indices gives $z_{abc} = z_{bac}$, and $\delta(z) = 0$ gives $z_{abc} = z_{acb}$. These transpositions generate every permutation of the three indices, so z_{abc} is fully symmetric. Thus $z = \iota(g)$ for a unique $g \in \text{Sym}^3 V$, and

$$\ker \delta = \iota(\text{Sym}^3 V).$$

Intersecting with $E \otimes V$ says precisely that all first derivatives of g lie in E , which gives $E^{(1)}$. Rank-nullity and $\dim V = 9$ give the formula. $\square$

1.3. The flattening rank of the permanent. Write $E := E_{\text{perm}_3}$. Differentiation with respect to x_{ij} produces the 2×2 subpermanent obtained by deleting row i and column j . Therefore

$$E = \text{span}\{x_{ia}x_{jb} + x_{ib}x_{ja} : 0 \leq i < j \leq 2, 0 \leq a < b \leq 2\}. \quad (3)$$

Distinct row-pair and column-pair supports are disjoint, so these nine quadrics are linearly independent:

$$\dim E = \binom{3}{2}^2 = 9. \quad (4)$$

Proposition 1.4.

$$E^{(1)} = \text{span}(\text{perm}_3).$$

In particular, $\dim E^{(1)} = 1$.

Proof. Every first derivative of perm_3 lies in E , so $\text{perm}_3 \in E^{(1)}$.

Conversely, let $g \in E^{(1)}$. Every quadratic monomial in (3) uses distinct variable rows and columns. Thus $\partial g / \partial x_{ij} \in E$ forces every nonzero cubic monomial of g to use three distinct rows and columns. Hence

$$g = \sum_{\sigma \in S_3} c_\sigma x_{0,\sigma(0)} x_{1,\sigma(1)} x_{2,\sigma(2)}.$$

Fix one variable and differentiate with respect to it. On the remaining two rows and columns there are two matchings. The corresponding part of E is one-dimensional and generated by their sum, so the two coefficients are equal. Thus $c_\sigma = c_\tau$ whenever σ and τ differ by a transposition. Transpositions generate S_3 , so all coefficients are equal and g is a scalar multiple of perm_3 . $\square$

Corollary 1.5.

$$\text{rank } K(\text{perm}_3) = 80.$$

Proof. Lemma 1.3, (4), and Proposition 1.4 give

$$\text{rank } K(\text{perm}_3) = 9 \cdot 9 - 1 = 80. \quad \square$$

1.4. A bound for one Chow term.

Proposition 1.6. *For every cubic Chow term $T = \ell_1 \ell_2 \ell_3$,*

$$\text{rank } K(T) \leq 26.$$

Equality holds when the three factors are linearly independent.

Proof. Let $F := E_T$. Differentiating T gives

$$F \subseteq \text{span}\{\ell_1 \ell_2, \ell_1 \ell_3, \ell_2 \ell_3\}, \quad \dim F \leq 3.$$

If $T \neq 0$, then $T \in F^{(1)}$, so $\dim F^{(1)} \geq 1$. Lemma 1.3 gives

$$\text{rank } K(T) = 9 \dim F - \dim F^{(1)} \leq 9 \cdot 3 - 1 = 26.$$

The zero term is immediate.

If the factors are linearly independent, a $GL(V)$ -change of coordinates reduces to $T = x_1 x_2 x_3$. Then

$$F = \text{span}\{x_1 x_2, x_1 x_3, x_2 x_3\}, \quad F^{(1)} = \text{span}(x_1 x_2 x_3),$$

so the rank is $27 - 1 = 26$. $\square$

1.5. The lower bound and a four-term decomposition.

Theorem 1.7.

$$\text{ChowRank}(\text{perm}_3) \geq 4.$$

Proof. Suppose $\text{perm}_3 = T_1 + \cdots + T_r$ is a sum of r Chow terms. The flattening is linear in the polynomial, and matrix rank is subadditive. Corollary 1.5 and Proposition 1.6 therefore give

$$80 = \text{rank } K(\text{perm}_3) \leq \sum_{s=1}^r \text{rank } K(T_s) \leq 26r.$$

Hence

$$r \geq \left\lceil \frac{80}{26} \right\rceil = 4.$$

□

Glynn's identity for $n = 3$ gives

$$\begin{aligned} 4 \text{perm}_3 &= \prod_{j=0}^2 (x_{0j} + x_{1j} + x_{2j}) \\ &\quad - \prod_{j=0}^2 (x_{0j} + x_{1j} - x_{2j}) \\ &\quad - \prod_{j=0}^2 (x_{0j} - x_{1j} + x_{2j}) \\ &\quad + \prod_{j=0}^2 (x_{0j} - x_{1j} - x_{2j}). \end{aligned} \tag{5}$$

Each line on the right is a product of three linear forms, and its scalar and sign may be absorbed into one factor. Thus $\text{ChowRank}(\text{perm}_3) \leq 4$. Together with Theorem 1.7, this proves Theorem 1.2.

1.6. Independent computation. The script `perm3_exact_verification.py` independently constructs the integer matrices from (3) and (1). Running

`python perm3_exact_verification.py`

returns the following values.

Quantity	Value
$\dim E$	9
$\dim E^{(1)}$	1
$\text{rank } K(\text{perm}_3)$	80
Koszul rank of a term with independent factors	26
Flattening lower bound	4
Number of terms in Glynn's decomposition	4

The program finds nonzero integer minors of orders 80 and 26 modulo 1000003. The prolongation computations above give the matching characteristic-zero upper bounds. Thus the replay is an exact cross-check, not floating-point or probabilistic evidence. The proof of the theorem is already complete without the computation.

The lower bound above is proved over the complex numbers. Any decomposition over a characteristic-zero field remains a decomposition after scalar extension to the complex numbers, while Glynn's decomposition is defined over the prime field. The equality therefore holds over every field of characteristic zero.